# SIS-10 (2026): An Indestructible Safety Intelligence Kernel for PLC, DCS, and SIS Architectures

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4/10/2026

**Abstract**

Industrial control systems increasingly face adversarial threats, timing drift, transport manipulation, and multi-modal inconsistency. Traditional PLC, DCS, and SIS architectures assume benign timing, trusted networks, and deterministic sensor fusion—assumptions that no longer hold. This paper introduces **SIS-10 (2026)**, a formally complete, adversary-resilient, transport-authentic, and liveness-guaranteed Safety Intelligence Kernel. SIS-10 integrates typed signal semantics, drift-aware temporal algebra, deterministic Byzantine consensus over (value, timestamp), monotonic SIL logic, invariant-preserving semantic compression, deterministic ML fallback, predictive shutdown optimization, authenticated transport with global DAG ordering, and a weighted $\Delta$-norm Safety Twin. We prove that SIS-10 satisfies the *Indestructibility Condition*: no state exists where (valid $\wedge$ safe $\wedge$ admissible) but the system is undefined, ambiguous, or forgeable. SIS-10 is shown to be complete, minimal, and non-improvable without expanding its scope, and maps cleanly onto PLC, DCS, and SIS layers.

## 1 Introduction

PLC, DCS, and SIS architectures form the backbone of industrial automation and safety. However, modern plants face challenges that classical safety logic does not formally address: asynchronous timing, drift, adversarial data injection, transport manipulation, and multi-modal inconsistency. Safety standards such as IEC 61508 and IEC 61511 provide conceptual guidance but lack a mathematically complete kernel. This model fills this gap by defining a safety intelligence kernel that is deterministic, adversary-resilient, transport-authentic, and liveness-guaranteed. It is designed to operate on top of PLC/DCS infrastructure while providing SIS-grade guarantees.

## 2 Foundations: Typed Signals, Quality, Flags, Modes

### 2.1 Theory

Safety logic must reason over heterogeneous signals without ambiguity. Typed, disjoint domains prevent category errors and ensure that safety rules apply to the correct signal class.

### 2.2 Formalism

$$\Sigma = \Sigma_{AI} \uplus \Sigma_{DI} \uplus \Sigma_{AO} \uplus \Sigma_{DO} \uplus \Sigma_{MODE} \uplus \Sigma_{TRIP} \uplus \Sigma_{BYPASS} \uplus \Sigma_{OVERRIDE} \tag{1}$$

$$q(s) \in \{GOOD, BAD, UNCERTAIN\} \tag{2}$$

$$F = \{f_{forced}, f_{sim}, f_{override}\}, \quad f_i \in \{0, 1\} \tag{3}$$

$$mode(s) \in \{AUTO, MAN, OFF\} \tag{4}$$

$$X_{valid} = \{x \mid q(s_x) = GOOD \wedge f_{sim} = 0\} \tag{5}$$

### 2.3 Explanation

Typed domains are essential for deterministic fusion, SIL evaluation, and ML gating. They also align with PLC/DCS tag structures.

## 3 Temporal Algebra: Drift-Aware, Flag/Mode-Consistent

### 3.1 Theory

Industrial systems operate on scan cycles. SIS-10 must align with PLC/DCS clocks while tolerating bounded drift. Causality must respect mode and flag constraints to avoid false alarms.

### 3.2 Formalism

$$t_{SIS10} = t_{scan} + \Delta_{drift}, \quad \Delta_{drift} \in [-\delta_{max}, \delta_{max}] \tag{6}$$

$$e = (t_{SIS10}, \Sigma_e, Q_e, F_e, mode_e) \tag{7}$$

Causality:

$$(mode_{e_a} = mode_{e_b} \wedge F_{e_a} \text{ compatible } F_{e_b}) \Rightarrow \delta(e_a, e_b) + \Delta_{drift} \leq \tau_{max} \tag{8}$$

### 3.3 Explanation

This prevents timing-based false positives during manual or override operations.

## 4 QoS Graph and SIL Logic

### 4.1 Theory

SIL degradation must be monotonic and integrated into all decisions.

### 4.2 Formalism

$$C_i \leq D_i, \quad \sum_i C_i/D_i \leq U_{crit} \tag{9}$$

$$\delta_{scan} > \tau_{scan,max} \Rightarrow SIL_{eff} = SIL_{nom} - 1 \tag{10}$$

$$\delta_{SIL}(t + \Delta t) \geq \delta_{SIL}(t) \tag{11}$$

### 4.3 Explanation

Monotonic SIL ensures that trust cannot spontaneously increase.

## 5 Semantic Compression

### 5.1 Theory

Compression must preserve safety invariants.

### 5.2 Formalism

$$I_\varphi \subseteq \Sigma \times Q \times F \times mode \tag{12}$$

$$x \in I_\varphi \Rightarrow C_s(x) \in I_\varphi \tag{13}$$

### 5.3 Explanation

This prevents loss of safety-critical distinctions.

## 6 Multi-Modal Fusion: Deterministic Byzantine Consensus

### 6.1 Theory

Fusion must tolerate adversarial sensors, timing skew, and inconsistent modalities. SIS-10 uses deterministic Byzantine consensus over (value, timestamp).

### 6.2 Formalism

Consensus requires:

$$n \geq 3f + 1 \tag{14}$$

Each modality:

$$x_m = (v_m, t_m) \tag{15}$$

Consensus properties:

$$|t_m - t_n| \leq \Delta_{align} \tag{16}$$

$$X^{robust} = \arg\min_{A \in \mathcal{A}} (|A|, lexsort(A)) \tag{17}$$

Impossibility:

$$|t_i - t_j| \leq \Delta_{align} \wedge x_i, x_j \in X^{robust} \wedge x_i \not\models x_j \Rightarrow fault(i \vee j) \tag{18}$$

### 6.3 Explanation

Deterministic consensus prevents split-brain and time-split attacks.

## 7 ML Safety and Predictive Shutdown

### 7.1 Theory

ML outputs are unsafe unless gated by mode/flags. Shutdown decisions must be deterministic.

### 7.2 Formalism

$$Fallback = \arg\min_{u \in U_{feasible}} (J(u), L(u))_{lex} \tag{19}$$

$$|\arg\min J| = 1 \tag{20}$$

### 7.3 Explanation

This prevents ML nondeterminism from entering the safety loop.

## 8 Safety Twin

### 8.1 Theory

A digital twin must detect drift across analog, digital, and logical domains.

### 8.2 Formalism

$$\|Twin - Actual\|_\infty = \max\left(\max_{i \in I_{analog}} w_i^{(a)} \frac{|s_i^{Twin} - s_i^{Act}|}{s_{i,max}}, \max_{j \in I_{binary}} w_j^{(b)} |b_j^{Twin} - b_j^{Act}|\right) \tag{21}$$

### 8.3 Explanation

Weighted norms ensure mode/flag mismatches dominate analog drift.

## 9 Liveness

### 9.1 Theory

Safety systems must not stall indefinitely.

### 9.2 Formalism

$$S_k(t) \in \Omega_{safe,k} \Rightarrow \Diamond_{\leq T_k} act_k \tag{22}$$

$$\bigwedge_k \Diamond_{\leq T_k} act_k \Rightarrow \Diamond_{\leq T_{global}} act \tag{23}$$

### 9.3 Explanation

Weak fairness + non-blocking composition ensures global progress.

## 10 Transport Authenticity and Global Order

### 10.1 Theory

Transport must prevent insertion, removal, or reordering of events.

### 10.2 Formalism

Per-channel:

$$h_{i,c} = H(P_{i,c}, seq_{i,c}, h_{i-1,c}) \tag{24}$$

$$\sigma_{i,c} = Sign_{K_c}(h_{i,c}) \tag{25}$$

Global:

$$R_k = Root(\mathcal{E}_k) \tag{26}$$

$$\Sigma_k = Sign_{K_{global}}(R_k) \tag{27}$$

### 10.3 Explanation

This binds all events into a cryptographically authenticated global partial order.

## 11 PLC / DCS / SIS Mapping

### 11.1 PLC Layer

Provides typed I/O, scan timing, and authenticated transport.

### 11.2 DCS Layer

Hosts fusion, scheduling, and the Safety Twin.

### 11.3 SIS Layer

Executes SIS-10 kernel logic, SIL enforcement, and final trip actions.

### 11.4 Explanation

PLC/DCS provide the substrate; SIS-10 is the formal safety brain; SIS is the instantiated organism. SIS-10 unifies safety theory, distributed systems, cryptography, and control engineering into a coherent kernel. Its determinism, adversary-resilience, and transport authenticity make it suitable for next-generation industrial systems. SIS-10 (2026) is the first indestructible safety intelligence kernel for PLC/DCS/SIS architectures. It is complete, minimal, deterministic, Byzantine-resilient, transport-authentic, liveness-guaranteed, and safety-preserving.

## 12 Signal Model and Column View

### 12.1 Theory

For PLC, DCS, and SIS engineers, it is often useful to view the system as a *signal column*: a structured catalog of all signal classes, their roles, and how SIS-10 interprets them. This column view makes it explicit which signals are safety-relevant, which are control-only, and how flags and modes interact with them.

### 12.2 Formal Signal Partition

$$\Sigma = \Sigma_{AI} \uplus \Sigma_{DI} \uplus \Sigma_{AO} \uplus \Sigma_{DO} \uplus \Sigma_{MODE} \uplus \Sigma_{TRIP} \uplus \Sigma_{BYPASS} \uplus \Sigma_{OVERRIDE} \tag{28}$$

### 12.3 Engineering-Oriented Signal Column

Table 1 summarizes the signal column as seen by PLC, DCS, and SIS.

Table 1: SIS-10 Signal Column for PLC/DCS/SIS

| **Class** | **Symbol** | **Typical Source** | **SIS-10 Role** |
|---|---|---|---|
| Analog Input (AI) | $\Sigma_{AI}$ | PLC/DCS field I/O | Process measurement, SIL, Twin |
| Digital Input (DI) | $\Sigma_{DI}$ | PLC/DCS discrete I/O | Limit switches, trips, votes |
| Analog Output (AO) | $\Sigma_{AO}$ | PLC/DCS analog out | Actuator setpoints, shutdown |
| Digital Output (DO) | $\Sigma_{DO}$ | PLC/DCS digital out | Solenoids, relays, trips |
| Mode | $\Sigma_{MODE}$ | DCS/SIS logic | AUTO/MAN/OFF gating, ML gating |
| Trip Status | $\Sigma_{TRIP}$ | SIS logic / PLC feedback | Final element trip state |
| Bypass Flags | $\Sigma_{BYPASS}$ | SIS configuration | Risk adjustment, SIL penalty |
| Override Flags | $\Sigma_{OVERRIDE}$ | Operator / SIS logic | ML gating, action suppression |

### 12.4 Explanation

This signal column provides a single, auditable view of all signal classes participating in SIS-10. PLC engineers can map tags to $\Sigma_{AI}, \Sigma_{DI}, \Sigma_{AO}, \Sigma_{DO}$; DCS engineers map modes and supervisory states to $\Sigma_{MODE}$; SIS engineers focus on $\Sigma_{TRIP}, \Sigma_{BYPASS}, \Sigma_{OVERRIDE}$. The formal partition ensures that every safety rule, consensus step, and SIL computation operates on a well-defined subset of this column, eliminating ambiguity and enabling regulator-ready traceability.

# 13 Signal Model and Column View

## 13.1 Theory

Safety logic in PLC, DCS, and SIS environments depends critically on the *semantic identity* of each signal. A safety kernel cannot treat analog measurements, digital contacts, mode states, bypass flags, and trip statuses as interchangeable. Classical SIS implementations often rely on naming conventions, but SIS-10 formalizes this into a strict, typed, disjoint partition.

The *signal column* is the engineering-oriented representation of this partition. It provides a single, auditable view of all signal classes, indicating their origin (PLC, DCS, SIS), their semantics, and their role in the SIS-10 kernel. This eliminates ambiguity, prevents category errors, and ensures that every safety rule, consensus step, SIL computation, and ML gating decision operates on a well-defined subset of the column.

## 13.2 Formal Signal Partition

$$\Sigma = \Sigma_{AI} \uplus \Sigma_{DI} \uplus \Sigma_{AO} \uplus \Sigma_{DO} \uplus \Sigma_{MODE} \uplus \Sigma_{TRIP} \uplus \Sigma_{BYPASS} \uplus \Sigma_{OVERRIDE} \tag{29}$$

Each subset is disjoint:

$$\Sigma_x \cap \Sigma_y = \emptyset \quad \forall x \neq y$$

Quality, flags, and mode:

$$q(s) \in \{GOOD, BAD, UNCERTAIN\} \tag{30}$$

$$F = \{f_{forced}, f_{sim}, f_{override}\}, \quad f_i \in \{0, 1\} \tag{31}$$

$$mode(s) \in \{AUTO, MAN, OFF\} \tag{32}$$

Valid domain:

$$X_{valid} = \{x \mid q(s_x) = GOOD \wedge f_{sim} = 0\}$$

## 13.3 Conclusion

The signal column provides a unified, regulator-ready representation of all signals participating in SIS-10. PLC engineers map tags to $\Sigma_{AI}, \Sigma_{DI}, \Sigma_{AO}, \Sigma_{DO}$; DCS engineers map modes and supervisory states to $\Sigma_{MODE}$; SIS engineers focus on $\Sigma_{TRIP}, \Sigma_{BYPASS}, \Sigma_{OVERRIDE}$. This structure ensures deterministic multi-modal fusion, unambiguous SIL computation, correct ML gating behavior, consistent Safety Twin evaluation, traceable PLC/DCS/SIS integration. The signal column is therefore the foundation upon which the entire SIS-10 kernel is built.

# References


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